25901321 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEHSC200 - FOOD SCIENCE AND QUALITY CONTROL

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is meant by loss of nutrients during cooking? [K] / [CO1]
2. Give one example of a commercially available probiotic drink. [K] / [CO3]
3. Mention the causes of food spoilage. [K] / [CO4]
4. How do the eyes of fresh fish appear? [K] / [CO2]
5. Which alternative protein source is obtained from microorganisms? [K] / [CO3]
6. State the objectives of GMP. [U] / [CO5]
7. Which sugar stage is suitable for brittle preparation? [K] / [CO1]
8. What is the role of diet in gene expression according to nutrigenomics? [K] / [CO3]
9. Give any two examples of intentional adulteration in foods. [K] / [CO4]
10. Name the main parts of an egg. [K] / [CO2]
11. Mention one application of AI in dairy processing. [U] / [CO3]

12. What is the substrate involved in enzymatic browning? [U] / [CO1]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. What is homogenization of milk? Explain the process and give two advantages. [K] / [CO2]

14. Write short notes on two applications of biotechnology in food processing with suitable examples. [K] / [CO3]

15. Describe the structure and main elements of the ISO 22000 food safety management system. [U] / [CO5]

16. Write short notes on the nutritional value of microgreens. [K] / [CO1]

17. Briefly discuss food-borne viral infections and their transmission through foods. [U] / [CO4]

18. Explain egg white foam formation – mention the stages of foam formation and any two factors affecting foam stability. [K] / [CO2]

19. Write short notes on anti-nutritional factors in pulses – trypsin inhibitors and lathyrogens (sources, effects and control measures). [K] / [CO1]

[4x5 = 20]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

20. Explain the salient features and objectives of the Food Safety and Standards Bill, 2005. [U] / [CO5]

21. Discuss the physiological and biochemical changes occurring in fruits during ripening. [U] / [CO1]

[10x1 = 10]